



Natural Language Processing

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Natural Language Processing

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Part 1: Introducing Hidden Markov Models

Modelling pairs of sequences

Input: sequence of words; Output: sequence of labels

Input	British	left	waffles	on	Falkland	Islands
Output1	N	N	V	P	N	N
Output2	N	V	N	P	N	N

⋮

N Noun, e.g. islands

V Verb, e.g. leave, left

P Preposition, e.g. on

Modelling pairs of sequences

Input: sequence of words; Output: sequence of labels

Input	British	left	waffles	on	Falkland	Islands
Output1	N	N	V	P	N	N
Output2	N	V	N	P	N	N
:						

- ▶ 3 states: $\mathcal{S} = \{N, V, P\}$
- ▶ Input sequence: $x_1, x_2, \dots, x_n$
- ▶ Output sequence: $t_1, t_2, \dots, t_n$ where $t_i \in \mathcal{S}$
- ▶ How many output sequences?

$$|\mathcal{S}|^n$$

Modelling pairs of sequences

Input: sequence of characters; Output: sequence of labels

Input 北京大学生比赛 7 chars

Output1 BIBIIBI 7 labels

Output2 BIIIBBI 7 labels

⋮ 7 labels

B Begin word

I Inside word

BIBIIBI 北京—大学生—比赛 (Beijing student competition)

BIIIBBI 北京大学—生—比赛 (Peking University Health
Competition)

Hidden Markov Models

- ▶ Input: x
- ▶ Output space: $\mathcal{Y}(x)$
- ▶ Output: $y \in \mathcal{Y}(x)$
- ▶ We want to learn a function f such that $f(x) = y$

Hidden Markov Models

Conditional model

- ▶ Construct function f using a conditional probability:

$$f(x) = \arg \max_{y \in \mathcal{Y}(x)} p(y | x)$$

- ▶ We can construct this function f using two principles:
 - ▶ Discriminative learning: find the best output y given input x
 - ▶ Generative modelling: model the joint probability $p(x, y)$ to find $p(y | x)$

Hidden Markov Models

Generative Model

- ▶ Start from the joint probability $p(x, y)$:

$$p(x, y) = p(y)p(x | y)$$

- ▶ Also:

$$p(x, y) = p(x)p(y | x)$$

Bayes Rule:

$$p(y | x) = \frac{p(y)p(x | y)}{p(x)}$$

Hidden Markov Models

Generative Model

- Bayes Rule:

$$p(y | x) = \frac{p(y)p(x | y)}{p(x)}$$

- where:

$$p(x) = \sum_{y \in \mathcal{Y}(x)} p(x, y) = \sum_{y \in \mathcal{Y}(x)} p(y)p(x | y)$$

- So using a generative model, we can find the best output y using:

$$p(y | x) = \frac{p(y)p(x | y)}{\sum_{y \in \mathcal{Y}(x)} p(y)p(x | y)}$$

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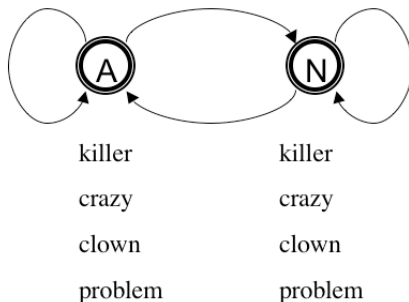
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Part 2: Algorithms for Hidden Markov Models

Hidden Markov Model

$$\text{Model } \theta = \begin{cases} \pi_i & p(i): \text{ starting at state } i \\ a_{i,j} & p(j | i): \text{ transition to state } i \text{ from state } j \\ b_i(o) & p(o | i): \text{ output } o \text{ at state } i \end{cases}$$



Hidden Markov Model Algorithms

- ▶ HMM as parser: compute the best sequence of states for a given observation sequence.
- ▶ HMM as language model: compute probability of given observation sequence.
- ▶ HMM as learner: given a corpus of observation sequences, learn its distribution, i.e. learn the parameters of the HMM from the corpus.
 - ▶ Learning from a set of observations with the sequence of states provided (states are not hidden) [\[Supervised Learning\]](#)
 - ▶ Learning from a set of observations without any state information. [\[Unsupervised Learning\]](#)

HMM as Parser


$$\pi =$$

A	0.25
N	0.75

$$a =$$

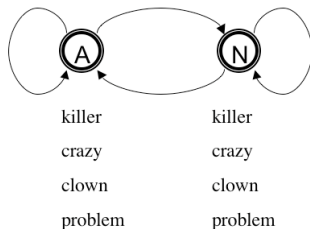
$a_{i,j}$	A	N
A	0.0	1.0
N	0.5	0.5

$$b =$$

$b_i(o)$	clown	killer	problem	crazy
A	0	0	0	1
N	0.4	0.3	0.3	0

The task: for a given observation sequence find the most likely state sequence. $a_{i,j} = p(j \mid i)$ and $b_i(o) = p(o \mid i)$

HMM as Parser



- ▶ Find most likely sequence of states for *killer clown*
- ▶ Score every possible sequence of states: AA, AN, NN, NA
 - ▶ $P(\text{killer clown, AA}) = \pi_A \cdot b_A(\text{killer}) \cdot a_{A,A} \cdot b_A(\text{clown}) = 0.0$
 - ▶ $P(\text{killer clown, AN}) = \pi_A \cdot b_A(\text{killer}) \cdot a_{A,N} \cdot b_N(\text{clown}) = 0.0$
 - ▶ $P(\text{killer clown, NN}) = \pi_N \cdot b_N(\text{killer}) \cdot a_{N,N} \cdot b_N(\text{clown}) = 0.75 \cdot 0.3 \cdot 0.5 \cdot 0.4 = 0.045$
 - ▶ $P(\text{killer clown, NA}) = \pi_N \cdot b_N(\text{killer}) \cdot a_{N,A} \cdot b_A(\text{clown}) = 0.0$
- ▶ Pick the state sequence with highest probability (NN=0.045).

HMM as Parser

- ▶ As we have seen, for input of length 2, and a HMM with 2 states there are 2^2 possible state sequences.
- ▶ In general, if we have q states and input of length T there are q^T possible state sequences.
- ▶ Using our example HMM, for input *killer crazy clown problem* we will have 2^4 possible state sequences to score.
- ▶ Our naive algorithm takes exponential time to find the best state sequence for a given input.
- ▶ The **Viterbi algorithm** uses dynamic programming to provide the best state sequence with a time complexity of $q^2 \cdot T$

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Part 3: Viterbi Algorithm for HMMs

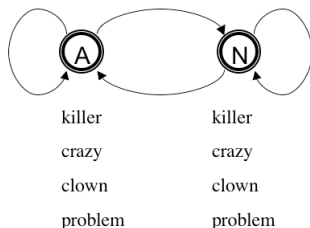
Viterbi Algorithm for HMMs

- ▶ For input of length T : $o_1, \dots, o_T$, we want to find the sequence of states $s_1, \dots, s_T$
- ▶ Each s_t in this sequence is one of the states in the HMM.
- ▶ So the task is to find the most likely sequence of states:

$$\arg \max_{s_1, \dots, s_T} P(o_1, \dots, o_T, s_1, \dots, s_T)$$

- ▶ The Viterbi algorithm solves this by creating a table $V[s, t]$ where s is one of the states, and t is an index between $1, \dots, T$.

Viterbi Algorithm for HMMs



- ▶ Consider the input *killer crazy clown problem*
- ▶ So the task is to find the most likely sequence of states:

$$\arg \max_{s_1, s_2, s_3, s_4} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, s_4)$$

- ▶ A sub-problem is to find the most likely sequence of states for *killer crazy clown*:

$$\arg \max_{s_1, s_2, s_3} P(\textit{killer crazy clown}, s_1, s_2, s_3)$$

Viterbi Algorithm for HMMs

- In our example there are two possible values for s_4 :

$$\begin{aligned} \max_{s_1, \dots, s_4} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, s_4) = \\ \max \left\{ \begin{aligned} &\max_{s_1, s_2, s_3} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, N), \\ &\max_{s_1, s_2, s_3} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, A) \end{aligned} \right\} \end{aligned}$$

- Similarly:

$$\begin{aligned} \max_{s_1, \dots, s_3} P(\textit{killer crazy clown}, s_1, s_2, s_3) = \\ \max \left\{ \begin{aligned} &\max_{s_1, s_2} P(\textit{killer crazy clown}, s_1, s_2, N), \\ &\max_{s_1, s_2} P(\textit{killer crazy clown}, s_1, s_2, A) \end{aligned} \right\} \end{aligned}$$

Viterbi Algorithm for HMMs

- ▶ Putting them together:

$$P(\text{killer crazy clown problem}, s_1, s_2, s_3, N) = \\ \max \{ P(\text{killer crazy clown}, s_1, s_2, N) \cdot a_{N,N} \cdot b_N(\text{problem}), \\ P(\text{killer crazy clown}, s_1, s_2, A) \cdot a_{A,N} \cdot b_N(\text{problem}) \}$$

$$P(\text{killer crazy clown problem}, s_1, s_2, s_3, A) = \\ \max \{ P(\text{killer crazy clown}, s_1, s_2, N) \cdot a_{N,A} \cdot b_A(\text{problem}), \\ P(\text{killer crazy clown}, s_1, s_2, A) \cdot a_{A,A} \cdot b_A(\text{problem}) \}$$

- ▶ The best score is given by:

$$\max_{s_1, \dots, s_4} P(\text{killer crazy clown problem}, s_1, s_2, s_3, s_4) = \\ \max_{N,A} \left\{ \max_{s_1, s_2, s_3} P(\text{killer crazy clown problem}, s_1, s_2, s_3, N), \right. \\ \left. \max_{s_1, s_2, s_3} P(\text{killer crazy clown problem}, s_1, s_2, s_3, A) \right\}$$

Viterbi Algorithm for HMMs

- Provide an index for each input symbol:

1:killer 2:crazy 3:clown 4:problem

$$V[N, 3] = \max_{s_1, s_2} P(\textit{killer crazy clown}, s_1, s_2, N)$$

$$V[N, 4] = \max_{s_1, s_2, s_3} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, N)$$

- Putting them together:

$$V[N, 4] = \max \{ V[N, 3] \cdot a_{N,N} \cdot b_N(\textit{problem}), \\ V[A, 3] \cdot a_{A,N} \cdot b_N(\textit{problem}) \}$$

$$V[A, 4] = \max \{ V[N, 3] \cdot a_{N,A} \cdot b_A(\textit{problem}), \\ V[A, 3] \cdot a_{A,A} \cdot b_A(\textit{problem}) \}$$

- The best score for the input is given by:
 $\max \{ V[N, 4], V[A, 4] \}$
- To extract the best sequence of states we backtrack (same trick as obtaining alignments from minimum edit distance)

Viterbi Algorithm for HMMs

- ▶ For input of length T : $o_1, \dots, o_T$, we want to find the sequence of states $s_1, \dots, s_T$
- ▶ Each s_t in this sequence is one of the states in the HMM.
- ▶ For each state q we initialize our table: $V[q, 1] = \pi_q \cdot b_q(o_1)$
- ▶ Then compute for $t = 1 \dots T - 1$ for each state q :

$$V[q, t + 1] = \max_{q'} \{ V[q', t] \cdot a_{q', q} \cdot b_q(o_{t+1}) \}$$

- ▶ After the loop terminates, the best score is $\max_q V[q, T]$

Learning from Fully Observed Data

$$\pi =$$

<i>A</i>	0.25
<i>N</i>	0.75

$$a =$$

$a_{i,j}$	<i>A</i>	<i>N</i>
<i>A</i>	0.0	1.0
<i>N</i>	0.5	0.5

$$b =$$

$b_i(o)$	<i>clown</i>	<i>killer</i>	<i>problem</i>	<i>crazy</i>
<i>A</i>	0	0	0	1
<i>N</i>	0.4	0.3	0.3	0

Viterbi algorithm:

V	killer:1	crazy:2	clown:3	problem:4
<i>A</i>				
<i>N</i>				

Learning from Fully Observed Data

$$\pi =$$

<i>A</i>	0.25
<i>N</i>	0.75

$$a =$$

$a_{i,j}$	<i>A</i>	<i>N</i>
<i>A</i>	0.0	1.0
<i>N</i>	0.5	0.5

$$b =$$

$b_i(o)$	<i>clown</i>	<i>killer</i>	<i>problem</i>	<i>crazy</i>
<i>A</i>	0	0	0	1
<i>N</i>	0.4	0.3	0.3	0

Viterbi algorithm:

V	killer:1	crazy:2	clown:3	problem:4
A	0	0.1125	0	0
N	0.225	0	0.045	0.00675

Probability models of language

Question

$\pi =$	V	0.25
	N	0.75

$a =$	$a_{i,j}$	V	N
	V	0.5	0.5
	N	0.5	0.5

$b =$	$b_i(o)$	<i>time</i>	<i>flies</i>	<i>can</i>
	V	0.1	0.1	0.8
	N	0.5	0.4	0.1

What is the best sequence of tags for each string below:

1. *time*
2. *time flies*
3. *time flies can*

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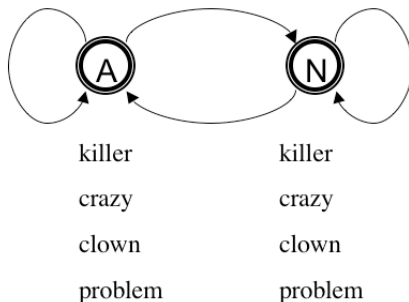
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Part 4: HMM as a Language Model

Hidden Markov Model

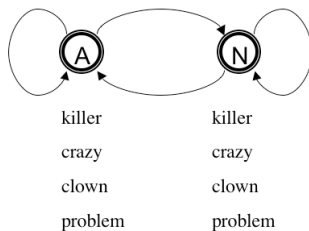
$$\text{Model } \theta = \begin{cases} \pi_i & p(i): \text{ starting at state } i \\ a_{i,j} & p(j | i): \text{ transition to state } i \text{ from state } j \\ b_i(o) & p(o | i): \text{ output } o \text{ at state } i \end{cases}$$



Hidden Markov Model Algorithms

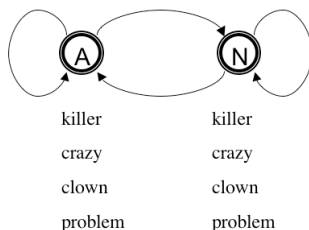
- ▶ HMM as parser: compute the best sequence of states for a given observation sequence.
- ▶ HMM as language model: compute probability of given observation sequence.
- ▶ HMM as learner: given a corpus of observation sequences, learn its distribution, i.e. learn the parameters of the HMM from the corpus.
 - ▶ Learning from a set of observations with the sequence of states provided (states are not hidden) [\[Supervised Learning\]](#)
 - ▶ Learning from a set of observations without any state information. [\[Unsupervised Learning\]](#)

HMM as a Language Model



- Find $P(\text{killer clown}) = \sum_y P(y, \text{killer clown})$
- $P(\text{killer clown}) = P(AA, \text{killer clown}) + P(AN, \text{killer clown}) + P(NN, \text{killer clown}) + P(NA, \text{killer clown})$

HMM as a Language Model



- ▶ Consider the input *killer crazy clown problem*
- ▶ So the task is to find the sum over all sequences of states:

$$\sum_{s_1, s_2, s_3, s_4} P(\text{killer crazy clown problem}, s_1, s_2, s_3, s_4)$$

- ▶ A sub-problem is to find the most likely sequence of states for *killer crazy clown*:

$$\sum_{s_1, s_2, s_3} P(\text{killer crazy clown}, s_1, s_2, s_3)$$

HMM as a Language Model

- In our example there are two possible values for s_4 :

$$\begin{aligned} \sum_{s_1, \dots, s_4} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, s_4) = \\ \sum_{s_1, s_2, s_3} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, N) + \\ \sum_{s_1, s_2, s_3} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, A) \end{aligned}$$

- Very similar to the Viterbi algorithm. Sum instead of max, and that's the only difference!

HMM as a Language Model

- Provide an index for each input symbol:

1:killer 2:crazy 3:clown 4:problem

$$V[N, 3] = \sum_{s_1, s_2} P(\textit{killer crazy clown}, s_1, s_2, N)$$

$$V[N, 4] = \sum_{s_1, s_2, s_3} P(\textit{killer crazy clown problem}, s_1, s_2, s_3, N)$$

- Putting them together:

$$V[N, 4] = V[N, 3] \cdot a_{N,N} \cdot b_N(\textit{problem}) + \\ V[A, 3] \cdot a_{A,N} \cdot b_N(\textit{problem})$$

$$V[A, 4] = V[N, 3] \cdot a_{N,A} \cdot b_A(\textit{problem}) + \\ V[A, 3] \cdot a_{A,A} \cdot b_A(\textit{problem})$$

- The best score for the input is given by: $V[N, 4] + V[A, 4]$

HMM as a Language Model

- ▶ For input of length T : $o_1, \dots, o_T$, we want to find $P(o_1, \dots, o_T) = \sum_{y_1, \dots, y_T} P(y_1, \dots, y_T, o_1, \dots, o_T)$
- ▶ Each y_t in this sequence is one of the states in the HMM.
- ▶ For each state q we initialize our table: $V[q, 1] = \pi_q \cdot b_q(o_1)$
- ▶ Then compute recursively for $t = 1 \dots T - 1$ for each state q :

$$V[q, t + 1] = \sum_{q'} \{ V[q', t] \cdot a_{q', q} \cdot b_q(o_{t+1}) \}$$

- ▶ After the loop terminates, the best score is $\sum_q V[q, T]$
- ▶ So: Viterbi with sum instead of max gives us an algorithm for HMM as a language model.
- ▶ This algorithm is sometimes called the *forward algorithm*.

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Part 5: Supervised Learning for HMMs

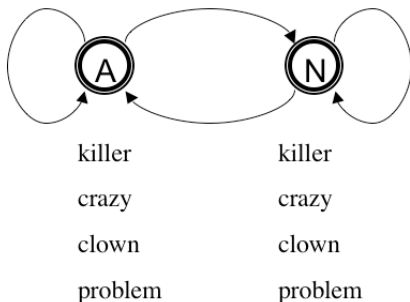
Hidden Markov Model Algorithms

- ▶ HMM as parser: compute the best sequence of states for a given observation sequence.
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- ▶ HMM as learner: given a corpus of observation sequences, learn its distribution, i.e. learn the parameters of the HMM from the corpus.
 - ▶ Learning from a set of observations with the sequence of states provided (states are not hidden) [\[Supervised Learning\]](#)
 - ▶ Learning from a set of observations without any state information. [\[Unsupervised Learning\]](#)

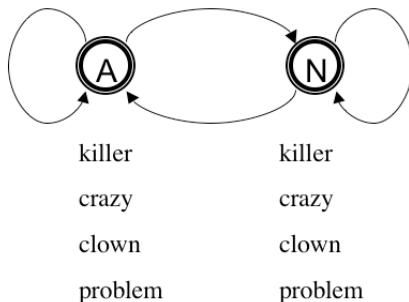
Hidden Markov Model

$$\text{Model } \theta = \begin{cases} \pi_i & \text{probability of starting at state } i \\ a_{i,j} & \text{probability of transition from state } i \text{ to state } j \\ b_i(o) & \text{probability of output } o \text{ at state } i \end{cases}$$

$$\text{Constraints : } \sum_i \pi_i = 1, \sum_j a_{i,j} = 1, \sum_o b_i(o) = 1$$



HMM Learning from Labeled Data



- ▶ The task: to find the values for the parameters of the HMM:
 - ▶ π_A, π_N
 - ▶ $a_{A,A}, a_{A,N}, a_{N,N}, a_{N,A}$
 - ▶ $b_A(killer), b_A(crazy), b_A(clown), b_A(problem)$
 - ▶ $b_N(killer), b_N(crazy), b_N(clown), b_N(problem)$

Learning from Fully Observed Data

Labeled Data L

x1,y1: killer/N clown/N (x1 = killer,clown; y1 = N,N)
x2,y2: killer/N problem/N (x2 = killer,problem; y2 = N,N)
x3,y3: crazy/A problem/N ...
x4,y4: crazy/A clown/N
x5,y5: problem/N crazy/A clown/N
x6,y6: clown/N crazy/A killer/N

Learning from Fully Observed Data

- ▶ Let's say we have m labeled examples:

$$L = (x_1, y_1), \dots, (x_m, y_m)$$

- ▶ Each $(x_\ell, y_\ell) = \{o_1, \dots, o_T, s_1, \dots, s_T\}$

- ▶ For each (x_ℓ, y_ℓ) we can compute the probability using the HMM:

- ▶ $(x_1 = \text{killer}, \text{clown}; y_1 = N, N) :$

$$P(x_1, y_1) = \pi_N \cdot b_N(\text{killer}) \cdot a_{N,N} \cdot b_N(\text{clown})$$

- ▶ $(x_2 = \text{killer}, \text{problem}; y_2 = N, N) :$

$$P(x_2, y_2) = \pi_N \cdot b_N(\text{killer}) \cdot a_{N,N} \cdot b_N(\text{problem})$$

- ▶ $(x_3 = \text{crazy}, \text{problem}; y_3 = A, N) :$

$$P(x_3, y_3) = \pi_A \cdot b_A(\text{crazy}) \cdot a_{A,N} \cdot b_N(\text{problem})$$

- ▶ $(x_4 = \text{crazy}, \text{clown}; y_4 = A, N) :$

$$P(x_4, y_4) = \pi_A \cdot b_A(\text{crazy}) \cdot a_{A,N} \cdot b_N(\text{clown})$$

- ▶ $(x_5 = \text{problem}, \text{crazy}, \text{clown}; y_5 = N, A, N) :$

$$P(x_5, y_5) = \pi_N \cdot b_N(\text{problem}) \cdot a_{N,A} \cdot b_A(\text{crazy}) \cdot a_{A,N} \cdot b_N(\text{clown})$$

- ▶ $(x_6 = \text{clown}, \text{crazy}, \text{killer}; y_6 = N, A, N) :$

$$P(x_6, y_6) = \pi_N \cdot b_N(\text{clown}) \cdot a_{N,A} \cdot b_A(\text{crazy}) \cdot a_{A,N} \cdot b_N(\text{killer})$$

- ▶ $\prod_\ell P(x_\ell, y_\ell) = \pi_N^4 \cdot \pi_A^2 \cdot a_{N,N}^2 \cdot a_{N,A}^2 \cdot a_{A,N}^4 \cdot a_{A,A}^0 \cdot b_N(\text{killer})^3 \cdot b_N(\text{clown})^4 \cdot b_N(\text{problem})^3 \cdot b_A(\text{crazy})^4$

Learning from Fully Observed Data

- ▶ We can easily collect frequency of observing a word with a state (tag)
 - ▶ $f(i, x, y)$ = number of times i is the initial state in (x, y)
 - ▶ $f(i, j, x, y)$ = number of times j follows i in (x, y)
 - ▶ $f(i, o, x, y)$ = number of times i is paired with observation o
- ▶ Then according to our HMM the probability of x, y is:

$$P(x, y) = \prod_i \pi_i^{f(i, x, y)} \cdot \prod_{i, j} a_{i, j}^{f(i, j, x, y)} \cdot \prod_{i, o} b_i(o)^{f(i, o, x, y)}$$

Learning from Fully Observed Data

- ▶ According to our HMM the probability of x, y is:

$$P(x, y) = \prod_i \pi_i^{f(i, x, y)} \cdot \prod_{i, j} a_{i, j}^{f(i, j, x, y)} \cdot \prod_{i, o} b_i(o)^{f(i, o, x, y)}$$

- ▶ For the labeled data $L = (x_1, y_1), \dots, (x_\ell, y_\ell), \dots, (x_m, y_m)$

$$\begin{aligned} P(L) &= \prod_{\ell=1}^m P(x_\ell, y_\ell) \\ &= \prod_{\ell=1}^m \left(\prod_i \pi_i^{f(i, x_\ell, y_\ell)} \cdot \prod_{i, j} a_{i, j}^{f(i, j, x_\ell, y_\ell)} \cdot \prod_{i, o} b_i(o)^{f(i, o, x_\ell, y_\ell)} \right) \end{aligned}$$

Learning from Fully Observed Data

- ▶ According to our HMM the probability of x, y is:

$$P(L) = \prod_{\ell=1}^m \left(\prod_i \pi_i^{f(i, x_\ell, y_\ell)} \cdot \prod_{i,j} a_{i,j}^{f(i,j, x_\ell, y_\ell)} \cdot \prod_{i,o} b_i(o)^{f(i,o, x_\ell, y_\ell)} \right)$$

- ▶ The log probability of the labeled data $(x_1, y_1), \dots, (x_m, y_m)$ according to HMM with parameters θ is:

$$\begin{aligned} L(\theta) &= \sum_{\ell=1}^m \log P(x_\ell, y_\ell) \\ &= \sum_{\ell=1}^m \sum_i f(i, x_\ell, y_\ell) \log \pi_i + \\ &\quad \sum_{i,j} f(i, j, x_\ell, y_\ell) \log a_{i,j} + \\ &\quad \sum_{i,o} f(i, o, x_\ell, y_\ell) \log b_i(o) \end{aligned}$$

Learning from Fully Observed Data

$$L(\theta) = \sum_{\ell=1}^m \sum_i f(i, x_{\ell}, y_{\ell}) \log \pi_i + \sum_{i,j} f(i, j, x_{\ell}, y_{\ell}) \log a_{i,j} + \sum_{i,o} f(i, o, x_{\ell}, y_{\ell}) \log b_i(o)$$

- ▶ $\theta = (\pi, a, b)$
- ▶ $L(\theta)$ is the log probability of the labeled data $(x_1, y_1), \dots, (x_m, y_m)$
- ▶ We want to find a θ that will give us the maximum value of $L(\theta)$
- ▶ Find the θ such that $\frac{dL(\theta)}{d\theta} = 0$

Learning from Fully Observed Data

$$L(\theta) = \sum_{\ell=1}^m \sum_i f(i, \mathbf{x}_\ell, y_\ell) \log \pi_i + \sum_{i,j} f(i, j, \mathbf{x}_\ell, y_\ell) \log a_{i,j} + \sum_{i,o} f(i, o, \mathbf{x}_\ell, y_\ell) \log b_i(o)$$

- The values of $\pi_i, a_{i,j}, b_i(o)$ that maximize $L(\theta)$ are:

$$\begin{aligned}\pi_i &= \frac{\sum_{\ell} f(i, \mathbf{x}_\ell, y_\ell)}{\sum_{\ell} \sum_k f(k, \mathbf{x}_\ell, y_\ell)} \\ a_{i,j} &= \frac{\sum_{\ell} f(i, j, \mathbf{x}_\ell, y_\ell)}{\sum_{\ell} \sum_k f(i, k, \mathbf{x}_\ell, y_\ell)} \\ b_i(o) &= \frac{\sum_{\ell} f(i, o, \mathbf{x}_\ell, y_\ell)}{\sum_{\ell} \sum_{o' \in V} f(i, o', \mathbf{x}_\ell, y_\ell)}\end{aligned}$$

Learning from Fully Observed Data

Labeled Data:

x1,y1: killer/N clown/N

x2,y2: killer/N problem/N

x3,y3: crazy/A problem/N

x4,y4: crazy/A clown/N

x5,y5: problem/N crazy/A clown/N

x6,y6: clown/N crazy/A killer/N

Learning from Fully Observed Data

- ▶ The values of π_i that maximize $L(\theta)$ are:

$$\pi_i = \frac{\sum_{\ell} f(i, \mathbf{x}_{\ell}, y_{\ell})}{\sum_{\ell} \sum_k f(k, \mathbf{x}_{\ell}, y_{\ell})}$$

- ▶ $\pi_N = \frac{2}{3}$ and $\pi_A = \frac{1}{3}$ because:

$$\sum_{\ell} f(N, \mathbf{x}_{\ell}, y_{\ell}) = 4$$

$$\sum_{\ell} f(A, \mathbf{x}_{\ell}, y_{\ell}) = 2$$

Learning from Fully Observed Data

- ▶ The values of $a_{i,j}$ that maximize $L(\theta)$ are:

$$a_{i,j} = \frac{\sum_{\ell} f(i, j, x_{\ell}, y_{\ell})}{\sum_{\ell} \sum_k f(i, k, x_{\ell}, y_{\ell})}$$

- ▶ $a_{N,N} = \frac{1}{2}$; $a_{N,A} = \frac{1}{2}$; $a_{A,N} = 1$ and $a_{A,A} = 0$ because:

$$\sum_{\ell} f(N, N, x_{\ell}, y_{\ell}) = 2$$

$$\sum_{\ell} f(N, A, x_{\ell}, y_{\ell}) = 2$$

$$\sum_{\ell} f(A, N, x_{\ell}, y_{\ell}) = 4$$

$$\sum_{\ell} f(A, A, x_{\ell}, y_{\ell}) = 0$$

Learning from Fully Observed Data

- ▶ The values of $b_i(o)$ that maximize $L(\theta)$ are:

$$b_i(o) = \frac{\sum_{\ell} f(i, o, x_{\ell}, y_{\ell})}{\sum_{\ell} \sum_{o' \in V} f(i, o', x_{\ell}, y_{\ell})}$$

- ▶ $b_N(killer) = \frac{3}{10}$; $b_N(clown) = \frac{4}{10}$; $b_N(problem) = \frac{3}{10}$ and $b_A(crazy) = 1$ because:

$$\sum_{\ell} f(N, killer, x_{\ell}, y_{\ell}) = 3$$

$$\sum_{\ell} f(A, killer, x_{\ell}, y_{\ell}) = 0$$

$$\sum_{\ell} f(N, clown, x_{\ell}, y_{\ell}) = 4$$

$$\sum_{\ell} f(A, clown, x_{\ell}, y_{\ell}) = 0$$

$$\sum_{\ell} f(N, crazy, x_{\ell}, y_{\ell}) = 0$$

$$\sum_{\ell} f(A, crazy, x_{\ell}, y_{\ell}) = 4$$

$$\sum_{\ell} f(N, problem, x_{\ell}, y_{\ell}) = 3$$

$$\sum_{\ell} f(A, problem, x_{\ell}, y_{\ell}) = 0$$

Learning from Fully Observed Data

x1,y1: killer/N clown/N

x2,y2: killer/N problem/N

x3,y3: crazy/A problem/N

x4,y4: crazy/A clown/N

x5,y5: problem/N crazy/A clown/N

x6,y6: clown/N crazy/A killer/N

$$\pi =$$

A	0.25
N	0.75

$$a =$$

$a_{i,j}$	A	N
A	0.0	1.0
N	0.5	0.5

$$b =$$

$b_i(o)$	clown	killer	problem	crazy
A	0	0	0	1
N	0.4	0.3	0.3	0

Natural Language Processing

Anoop Sarkar

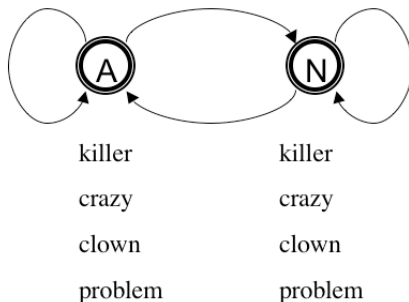
anoopsarkar.github.io/nlp-class

Simon Fraser University

Part 6: Lagrange Multipliers

Hidden Markov Model

$$\text{Model } \theta = \begin{cases} \pi_i & p(i): \text{ starting at state } i \\ a_{i,j} & p(j | i): \text{ transition to state } i \text{ from state } j \\ b_i(o) & p(o | i): \text{ output } o \text{ at state } i \end{cases}$$



Learning from Fully Observed Data

$$L(\theta) = \sum_{\ell=1}^m \sum_i f(i, x_{\ell}, y_{\ell}) \log \pi_i + \sum_{i,j} f(i, j, x_{\ell}, y_{\ell}) \log a_{i,j} + \sum_{i,o} f(i, o, x_{\ell}, y_{\ell}) \log b_i(o)$$

- ▶ $\theta = (\pi, a, b)$
- ▶ $L(\theta)$ is the log probability of the labeled data $(x_1, y_1), \dots, (x_m, y_m)$
- ▶ We want to find a θ that will give us the maximum value of $L(\theta)$
- ▶ Find the θ such that $\frac{dL(\theta)}{d\theta} = 0$

Learning from Fully Observed Data

$$L(\theta) = \sum_{\ell=1}^m \sum_i f(i, x_{\ell}, y_{\ell}) \log \pi_i + \sum_{i,j} f(i, j, x_{\ell}, y_{\ell}) \log a_{i,j} + \sum_{i,o} f(i, o, x_{\ell}, y_{\ell}) \log b_i(o)$$

- ▶ Find the θ such that $\frac{dL(\theta)}{d\theta} = 0$ and $\theta = (\pi, a, b)$
- ▶ Split up $L(\theta)$ into $L(\pi), L(a), L(b)$
- ▶ Let $\nabla L = \forall i, j, o : \frac{\partial L(\pi)}{\partial \pi_i}, \frac{\partial L(a)}{\partial a_{i,j}}, \frac{\partial L(b)}{\partial b_i(o)}$
- ▶ We must also obey constraints:
 $\sum_k \pi_k = 1, \sum_k a_{i,k} = 1, \sum_o b_i(o) = 1$

Learning from Fully Observed Data

$$L(\pi) = \sum_{\ell=1}^m \sum_i f(i, x_{\ell}, y_{\ell}) \log \pi_i$$

- ▶ Let us focus on $\nabla L(\pi)$ (the other two: a and b are similar)
- ▶ For the constraint $\sum_k \pi_k = 1$ we introduce a new variable into our search for a maximum:

$$L(\pi, \lambda) = L(\pi) + \lambda(1 - \sum_k \pi_k)$$

- ▶ λ is called the Lagrange multiplier
- ▶ λ penalizes any solution that does not obey the constraint
- ▶ The constraint ensures that π is a probability distribution

Learning from Fully Observed Data

$$\frac{\partial L(\pi)}{\partial \pi_i} = \frac{\partial}{\partial \pi_i} \underbrace{\sum_{\ell=1}^m f(i, x_\ell, y_\ell) \log \pi_i}_{\text{the only part with variable } \pi_i} + \underbrace{\sum_{\ell=1}^m \sum_{j:j \neq i} f(j, x_\ell, y_\ell) \log \pi_j}_{\text{no } \pi_i \text{ so derivative is 0}}$$

- We want a value of π_i such that $\frac{\partial L(\pi, \lambda)}{\partial \pi_i} = 0$

$$\frac{\partial}{\partial \pi_i} \sum_{\ell=1}^m \left(f(i, x_\ell, y_\ell) \log \pi_i + \lambda (1 - \sum_k \pi_k) \right) = 0$$
$$\frac{\partial}{\partial \pi_i} \sum_{\ell=1}^m \left(\underbrace{f(i, x_\ell, y_\ell) \log \pi_i}_{\frac{\partial}{\partial \pi_i} = \frac{f(i, x_\ell, y_\ell)}{\pi_i}} + \lambda - \underbrace{\lambda \pi_i}_{\frac{\partial}{\partial \pi_i} = \lambda} - \lambda \sum_{j:j \neq i} \pi_j \right) = 0$$

Learning from Fully Observed Data

$$\frac{\partial L(\pi)}{\partial \pi_i} = \frac{\partial}{\partial \pi_i} \underbrace{\sum_{\ell=1}^m f(i, x_{\ell}, y_{\ell}) \log \pi_i}_{\text{the only part with variable } \pi_i} + \underbrace{\sum_{\ell=1}^m \sum_{j:j \neq i} f(j, x_{\ell}, y_{\ell}) \log \pi_j}_{\text{no } \pi_i \text{ so derivative is 0}}$$

- We can obtain a value of π_i wrt λ :

$$\frac{\partial L(\pi, \lambda)}{\partial \pi_i} = \underbrace{\sum_{\ell=1}^m \frac{f(i, x_{\ell}, y_{\ell})}{\pi_i}}_{\text{see previous slide}} - \lambda = 0$$
$$\pi_i = \frac{\sum_{\ell=1}^m f(i, x_{\ell}, y_{\ell})}{\lambda} \quad (1)$$

- Combine π_i s from Eqn (1) with constraint $\sum_k \pi_k = 1$

$$\lambda = \sum_k \sum_{\ell=1}^m f(k, x_{\ell}, y_{\ell})$$

Learning from Fully Observed Data

$$\frac{\partial L(\pi)}{\partial \pi_i} = \frac{\partial}{\partial \pi_i} \underbrace{\sum_{\ell=1}^m f(i, x_{\ell}, y_{\ell}) \log \pi_i}_{\text{the only part with variable } \pi_i} + \underbrace{\sum_{\ell=1}^m \sum_{j:j \neq i} f(j, x_{\ell}, y_{\ell}) \log \pi_j}_{\text{no } \pi_i \text{ so derivative is 0}}$$

- The value of π_i for which $\frac{\partial L(\pi, \lambda)}{\partial \pi_i} = 0$ is Eqn (2) which can be combined with the value of λ from Eqn (3).

$$\pi_i = \frac{\sum_{\ell=1}^m f(i, x_{\ell}, y_{\ell})}{\lambda} \quad (2)$$

$$\lambda = \sum_k \sum_{\ell=1}^m f(k, x_{\ell}, y_{\ell}) \quad (3)$$

$$\pi_i = \frac{\sum_{\ell=1}^m f(i, x_{\ell}, y_{\ell})}{\sum_k \sum_{\ell=1}^m f(k, x_{\ell}, y_{\ell})}$$

Learning from Fully Observed Data

$$L(\theta) = \sum_{\ell=1}^m \sum_i f(i, \mathbf{x}_\ell, y_\ell) \log \pi_i + \sum_{i,j} f(i, j, \mathbf{x}_\ell, y_\ell) \log a_{i,j} + \sum_{i,o} f(i, o, \mathbf{x}_\ell, y_\ell) \log b_i(o)$$

- The values of $\pi_i, a_{i,j}, b_i(o)$ that maximize $L(\theta)$ are:

$$\begin{aligned}\pi_i &= \frac{\sum_{\ell} f(i, \mathbf{x}_\ell, y_\ell)}{\sum_{\ell} \sum_k f(k, \mathbf{x}_\ell, y_\ell)} \\ a_{i,j} &= \frac{\sum_{\ell} f(i, j, \mathbf{x}_\ell, y_\ell)}{\sum_{\ell} \sum_k f(i, k, \mathbf{x}_\ell, y_\ell)} \\ b_i(o) &= \frac{\sum_{\ell} f(i, o, \mathbf{x}_\ell, y_\ell)}{\sum_{\ell} \sum_{o' \in V} f(i, o', \mathbf{x}_\ell, y_\ell)}\end{aligned}$$

Natural Language Processing

Anoop Sarkar

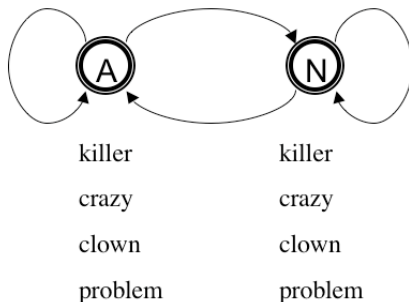
anoopsarkar.github.io/nlp-class

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Part 7: Unsupervised Learning for HMMs

Hidden Markov Model

$$\text{Model } \theta = \begin{cases} \pi_i & p(i): \text{ starting at state } i \\ a_{i,j} & p(j | i): \text{ transition to state } i \text{ from state } j \\ b_i(o) & p(o | i): \text{ output } o \text{ at state } i \end{cases}$$



Hidden Markov Model Algorithms

- ▶ HMM as parser: compute the best sequence of states for a given observation sequence.
- ▶ HMM as language model: compute probability of given observation sequence.
- ▶ HMM as learner: given a corpus of observation sequences, learn its distribution, i.e. learn the parameters of the HMM from the corpus.
 - ▶ Learning from a set of observations with the sequence of states provided (states are not hidden) [\[Supervised Learning\]](#)
 - ▶ Learning from a set of observations without any state information. [\[Unsupervised Learning\]](#)

Learning from Unlabeled Data

Unlabeled Data $U = x_1, \dots, x_m$:

x1: killer clown

x2: killer problem

x3: crazy problem

x4: crazy clown

- ▶ y_1, y_2, y_3, y_4 are unknown.
- ▶ But we can enumerate all possible values for y_1, y_2, y_3, y_4

- ▶ For example, for x1: killer clown

x1,y1,1:	killer/A clown/A	$p_1 = \pi_A \cdot b_A(killer) \cdot a_{A,A} \cdot b_A(clown)$
x1,y1,2:	killer/A clown/N	$p_2 = \pi_A \cdot b_A(killer) \cdot a_{A,N} \cdot b_N(clown)$
x1,y1,3:	killer/N clown/N	$p_3 = \pi_N \cdot b_N(killer) \cdot a_{N,N} \cdot b_N(clown)$
x1,y1,4:	killer/N clown/A	$p_4 = \pi_N \cdot b_N(killer) \cdot a_{N,A} \cdot b_A(clown)$

Learning from Unlabeled Data

- ▶ Assume some values for $\theta = \pi, a, b$
- ▶ We can compute $P(y \mid x_\ell, \theta)$ for any y for a given x_ℓ

$$P(y \mid x_\ell, \theta) = \frac{P(x, y \mid \theta)}{\sum_{y'} P(x, y' \mid \theta)}$$

- ▶ For example, we can compute $P(\text{NN} \mid \text{killer clown}, \theta)$ as follows:

$$\frac{\pi_N \cdot b_N(\text{killer}) \cdot a_{N,N} \cdot b_N(\text{clown})}{\sum_{i,j} \pi_i \cdot b_i(\text{killer}) \cdot a_{i,j} \cdot b_j(\text{clown})}$$

- ▶ $P(y \mid x_\ell, \theta)$ is called the *posterior probability*

Learning from Unlabeled Data

- ▶ Compute the posterior for all possible outputs for each example in training:
- ▶ For x_1 : killer clown
 - $x_1, y_1, 1$: killer/A clown/A $P(AA | \text{killer clown}, \theta)$
 - $x_1, y_1, 2$: killer/A clown/N $P(AN | \text{killer clown}, \theta)$
 - $x_1, y_1, 3$: killer/N clown/N $P(NN | \text{killer clown}, \theta)$
 - $x_1, y_1, 4$: killer/N clown/A $P(NA | \text{killer clown}, \theta)$
- ▶ For x_2 : killer problem
 - $x_2, y_2, 1$: killer/A problem/A $P(AA | \text{killer problem}, \theta)$
 - $x_2, y_2, 2$: killer/A problem/N $P(AN | \text{killer problem}, \theta)$
 - $x_2, y_2, 3$: killer/N problem/N $P(NN | \text{killer problem}, \theta)$
 - $x_2, y_2, 4$: killer/N problem/A $P(NA | \text{killer problem}, \theta)$
- ▶ Similarly for x_3 : crazy problem
- ▶ And x_4 : crazy clown

Learning from Unlabeled Data

- ▶ For unlabeled data, the log probability of the data given θ is:

$$\begin{aligned} L(\theta) &= \sum_{\ell=1}^m \log \sum_y P(x_\ell, y \mid \theta) \\ &= \sum_{\ell=1}^m \log \sum_y P(y \mid x_\ell, \theta) \cdot P(x_\ell \mid \theta) \end{aligned}$$

- ▶ Unlike the fully observed case there is no simple solution to finding θ to maximize $L(\theta)$
- ▶ We instead initialize θ to some values, and then iteratively find better values of θ : $\theta^0, \theta^1, \dots$ using the following formula:

$$\begin{aligned} \theta^t &= \arg \max_{\theta} Q(\theta, \theta^{t-1}) \\ &= \sum_{\ell=1}^m \sum_y P(y \mid x_\ell, \theta^{t-1}) \cdot \log P(x_\ell, y \mid \theta) \end{aligned}$$

Learning from Unlabeled Data

$$\begin{aligned}\theta^t &= \arg \max_{\theta} Q(\theta, \theta^{t-1}) \\ Q(\theta, \theta^{t-1}) &= \sum_{\ell=1}^m \sum_y P(y \mid x_{\ell}, \theta^{t-1}) \cdot \log P(x_{\ell}, y \mid \theta) \\ &= \sum_{\ell=1}^m \sum_y P(y \mid x_{\ell}, \theta^{t-1}) \cdot \\ &\quad \left(\sum_i f(i, x_{\ell}, y) \cdot \log \pi_i \right. \\ &\quad + \sum_{i,j} f(i, j, x_{\ell}, y) \cdot \log a_{i,j} \\ &\quad \left. + \sum_{i,o} f(i, o, x_{\ell}, y) \cdot \log b_i(o) \right)\end{aligned}$$

Learning from Unlabeled Data

$$g(i, x_\ell) = \sum_y P(y \mid x_\ell, \theta^{t-1}) \cdot f(i, x_\ell, y)$$

$$g(i, j, x_\ell) = \sum_y P(y \mid x_\ell, \theta^{t-1}) \cdot f(i, j, x_\ell, y)$$

$$g(i, o, x_\ell) = \sum_y P(y \mid x_\ell, \theta^{t-1}) \cdot f(i, o, x_\ell, y)$$

$$\begin{aligned} \theta^t = \arg \max_{\pi, a, b} & \sum_{\ell=1}^m \sum_i g(i, x_\ell) \cdot \log \pi_i \\ & + \sum_{i,j} g(i, j, x_\ell) \cdot \log a_{i,j} \\ & + \sum_{i,o} g(i, o, x_\ell) \cdot \log b_j(o) \end{aligned}$$

Learning from Unlabeled Data

$$Q(\theta, \theta^{t-1}) = \sum_{\ell=1}^m \sum_i g(i, x_\ell) \log \pi_i + \sum_{i,j} g(i, j, x_\ell) \log a_{i,j} + \sum_{i,o} g(i, o, x_\ell) \log b_i(o)$$

- The values of $\pi_i, a_{i,j}, b_i(o)$ that maximize $L(\theta)$ are:

$$\begin{aligned} \pi_i &= \frac{\sum_{\ell} g(i, x_{\ell})}{\sum_{\ell} \sum_k g(k, x_{\ell})} \\ a_{i,j} &= \frac{\sum_{\ell} g(i, j, x_{\ell})}{\sum_{\ell} \sum_k g(i, k, x_{\ell})} \\ b_i(o) &= \frac{\sum_{\ell} g(i, o, x_{\ell})}{\sum_{\ell} \sum_{o' \in V} g(i, o', x_{\ell})} \end{aligned}$$

EM Algorithm for Learning HMMs

- ▶ Initialize θ^0 at random. Let $t = 0$.
- ▶ The EM Algorithm:
 - ▶ E-step: compute expected values of y , $P(y \mid x, \theta)$ and calculate $g(i, x)$, $g(i, j, x)$, $g(i, o, x)$
 - ▶ M-step: compute $\theta^t = \arg \max_{\theta} Q(\theta, \theta^{t-1})$
 - ▶ Stop if $L(\theta^t)$ did not change much since last iteration. Else continue.
- ▶ The above algorithm is guaranteed to improve likelihood of the unlabeled data.
- ▶ In other words, $L(\theta^t) \geq L(\theta^{t-1})$
- ▶ *But* it all depends on θ^0 !

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